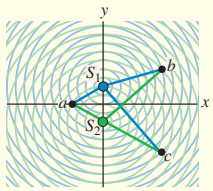
CHAPTER 35 SUMMARY

Interference and coherent sources: Monochromatic light is light with a single frequency. Coherence is a definite, unchanging phase relationship between two waves. The overlap of waves from two coherent sources of monochromatic light forms an interference pattern. The principle of superposition states that the total wave disturbance at any point is the sum of the disturbances from the separate waves.

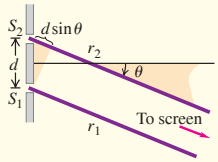


Two-source interference of light: When two sources are in phase, constructive interference occurs where the difference in path length from the two sources is zero or an integer number of wavelengths; destructive interference occurs where the path difference is a half-integer number of wavelengths. If two sources separated by a distance d are both very far from a point P, and the line from the sources to P makes an angle θ with the line perpendicular to the line of the sources, then the condition for constructive interference at P is Eq. (35.4). The condition for destructive interference is Eq. (35.5). When θ is very small, the position y_m of the mth bright fringe on a screen located a distance R from the sources is given by Eq. (35.6). (See Examples 35.1 and 35.2.)

$$d\sin\theta = m\lambda \quad (m = 0, \pm 1, \pm 2, \ldots) \quad (35.4)$$
(constructive interference)

$$d\sin\theta = \left(m + \tfrac{1}{2}\right)\lambda \quad (m = 0, \pm 1, \pm 2, \ldots) \quad (35.5)$$
(destructive interference)

$$y_m = R\frac{m\lambda}{d} \quad (m = 0, \pm 1, \pm 2, \ldots) \quad (35.6)$$
(bright fringes)

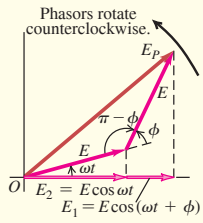


Intensity in interference patterns: When two sinusoidal waves with equal amplitude E and phase difference ϕ are superimposed, the resultant amplitude E_P and intensity I are given by Eqs. (35.7) and (35.10), respectively. If the two sources emit in phase, the phase difference ϕ at a point P (located a distance r_1 from source 1 and a distance r_2 from source 2) is directly proportional to the path difference $r_2 - r_1$. (See Example 35.3.)

$$E_P = 2E\left|\cos\frac{\phi}{2}\right| \quad (35.7)$$

$$I = I_0\cos^2\frac{\phi}{2} \quad (35.10)$$

$$\phi = \frac{2\pi}{\lambda}(r_2 - r_1) = k(r_2 - r_1) \quad (35.11)$$



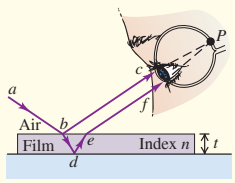
Interference in thin films: When light is reflected from both sides of a thin film of thickness t and no phase shift occurs at either surface, constructive interference between the reflected waves occurs when $2t$ is equal to an integer number of wavelengths. If a half-cycle phase shift occurs at one surface, this is the condition for destructive interference. A half-cycle phase shift occurs during reflection whenever the index of refraction in the second material is greater than that in the first. (See Examples 35.4–35.7.)

$$2t = m\lambda \quad (m = 0, 1, 2, \ldots) \quad (35.17a)$$
(constructive reflection from thin film, no relative phase shift)

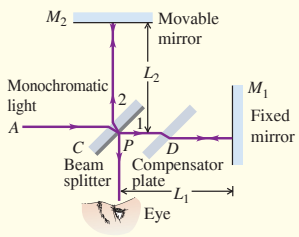
$$2t = \left(m + \tfrac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \ldots) \quad (35.17b)$$
(destructive reflection from thin film, no relative phase shift)

$$2t = \left(m + \tfrac{1}{2}\right)\lambda \quad (m = 0, 1, 2, \ldots) \quad (35.18a)$$
(constructive reflection from thin film, half-cycle phase shift)

$$2t = m\lambda \quad (m = 0, 1, 2, \ldots) \quad (35.18b)$$
(destructive reflection from thin film, half-cycle phase shift)



Michelson interferometer: The Michelson interferometer uses a monochromatic light source and can be used for high-precision measurements of wavelengths. Its original purpose was to detect motion of the earth relative to a hypothetical ether, the supposed medium for electromagnetic waves. The ether has never been detected, and the concept has been abandoned; the speed of light is the same relative to all observers. This is part of the foundation of the special theory of relativity.



Chapter 35 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review Examples EXAMPLES 35.1 and 35.2 (Section 35.2) before attempting these problems.

VP35.2.1 In a two-slit interference experiment like that shown in Fig. 35.7, the slits are 0.180 mm apart and the screen is 1.50 m from the slits. The $m = 2$ bright fringe is 11.4 mm from the central bright fringe. Find (a) the wavelength of the light and (b) the distance from the central bright fringe to the $m = -3$ bright fringe.

VP35.2.2 Two identical vertical radio antennas are 0.810 km apart. Both emit in phase at a frequency of $1.05\,\text{MHz} = 1.05 \times 10^6\,\text{Hz}$. Find the angles in the range from 0° and 90° at which the intensity measured several kilometers away from these antennas is (a) maximum and (b) minimum.

VP35.2.3 You shine red laser light of wavelength 685 nm on two closely spaced slits. This creates a pattern on a screen that is 2.10 m from the double slit. The $m = 3$ bright fringe in this pattern is 9.15 mm from the central bright fringe. (a) What is the distance between the slits? (b) What would be the distance between the central and $m = 3$ bright fringes if you used green laser light of wavelength 515 nm?

VP35.2.4 In a two-slit interference experiment the distance between the slits is 0.370 mm and the angle of the line from the slits to the $m = 4$ bright region on the screen is 0.407°. Find (a) the wavelength of the light used and (b) the angle of the line from the slits to the $m = 2$ dark region on the screen.

Be sure to review EXAMPLE 35.3 (Section 35.3) before attempting these problems.

VP35.3.1 Two identical radio antennas are 13.0 m apart and transmit in phase at frequency 59.3 MHz. At a distance of 0.850 km from the midpoint between the antennas in the direction $\theta = 0$, the intensity is $I_0 = 0.0330\ \text{W/m}^2$. Find (a) the intensity at this same distance in the direction $\theta = 5.00°$ and (b) the smallest angle for which the intensity is $I_0/2$.

VP35.3.2 A two-slit interference experiment uses laser light of wavelength 655 nm. The slits are 0.230 mm apart and 1.75 m from the screen on which the interference pattern appears. The intensity of the light at the central bright fringe is $I_0 = 0.0520\ \text{W/m}^2$. Find (a) the intensity at a point on the screen 6.50 mm from the central bright fringe and (b) the distance on the screen from the central bright fringe to the nearest point where the intensity is $I_0/4$.

VP35.3.3 You walk in a circle of radius 1.20 km around the midpoint of a pair of identical radio antennas that are 8.00 m apart. The intensity decreases continuously from $0.0540\ \text{W/m}^2$ in the direction $\theta = 0°$ to $0.0303\ \text{W/m}^2$ in the direction $\theta = 6.00°$. Find (a) the wavelength of the radio waves and (b) the intensity in the direction $\theta = 12.0°$.

VP35.3.4 A laser of unknown wavelength illuminates a pair of slits separated by 0.110 mm, producing an interference pattern on a wall 5.00 m from the slits. A point on the wall 10.5 mm from the central bright fringe is between that fringe and the $m = 1$ bright fringe, and the intensity there is 9.00% as great as at the central bright fringe. Find (a) the wavelength of the laser and (b) the distance from the central bright fringe to the nearest point on the wall where the intensity is 30.0% as great as at the central bright fringe.

Be sure to review EXAMPLES 35.4, 35.5, and 35.6 (Section 35.4) before attempting these problems.

VP35.6.1 Two identical microscope slides made of glass ($n = 1.50$) are stacked on top of each other. The slides are 8.00 cm long and are in contact at one end; at the other end they are separated by a piece of paper. You illuminate the slides from above with monochromatic light that has a wavelength in air of 565 nm. The spacing between the interference fringes seen by reflection is 1.30 mm. (a) Find the thickness of the piece of paper. (b) What would be the spacing between fringes if the space between the slides were filled with ethanol ($n = 1.36$) rather than air?

VP35.6.2 A thin, flat slide of glass ($n = 1.50$) lies on a table, and a second, identical glass slide is suspended 146 nm above the first one. The surfaces of the slides are parallel to each other. You illuminate the two slides from above. Find the wavelengths in air, if any, in the visible range (380 to 750 nm) for which there will be constructive reflection and destructive reflection if the space between the slides is filled with (a) air or (b) carbon tetrachloride ($n = 1.46$).

VP35.6.3 A thin, flat sheet of plastic ($n = 1.40$) lies on a table, and a second sheet of the same dimensions but made of flint glass ($n = 1.60$) is suspended 172 nm above the first sheet. The surfaces of the sheets are parallel to each other. You illuminate the two sheets from above. Find the wavelengths in air, if any, in the visible range (380 to 750 nm) for which there will be constructive reflection and destructive reflection if the space between the sheets is filled with (a) air or (b) ethanol ($n = 1.36$).

VP35.6.4 A bowl of water ($n = 1.33$) has a thin layer of grease ($n = 1.60$) floating on top of the water. When you illuminate the grease from above with monochromatic light, you get constructive reflection if the wavelength in air is 565 nm. Find (a) the smallest possible thickness of the grease layer and (b) the second smallest possible thickness.

BRIDGING PROBLEM Thin-Film Interference in an Oil Slick

An oil tanker spills a large amount of oil ($n = 1.45$) into the sea ($n = 1.33$). (a) If you look down onto the oil spill from overhead, what predominant wavelength of light do you see at a point where the oil is 380 nm thick? What color is the light? (*Hint:* See Table 32.1.) (b) In the water under the slick, what visible wavelength (as measured in air) is predominant in the transmitted light at the same place in the slick as in part (a)?

SOLUTION GUIDE

IDENTIFY and SET UP

1. The oil layer acts as a thin film, so we must consider interference between light that reflects from the top and bottom surfaces of the oil. If a wavelength is prominent in the *transmitted* light, there is destructive interference for that wavelength in the *reflected* light.
2. Choose the appropriate interference equations that relate the thickness of the oil film and the wavelength of light. Take account of the indexes of refraction of the air, oil, and water.

EXECUTE

3. For part (a), find the wavelengths for which there is constructive interference as seen from above the oil film. Which of these are in the visible spectrum?
4. For part (b), find the visible wavelength for which there is destructive interference as seen from above the film. (This will ensure that there is substantial transmitted light at the wavelength.)

EVALUATE

5. If a diver below the water's surface shines a light up at the bottom of the oil film, at what wavelengths would there be constructive interference in the light that reflects back downward?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q35.1 A two-slit interference experiment is set up, and the fringes are displayed on a screen. Then the whole apparatus is immersed in the nearest swimming pool. How does the fringe pattern change?

Q35.2 Could an experiment similar to Young's two-slit experiment be performed with sound? How might this be carried out? Does it matter that sound waves are longitudinal and electromagnetic waves are transverse? Explain.

Q35.3 Monochromatic coherent light passing through two thin slits is viewed on a distant screen. Are the bright fringes equally spaced on the screen? If so, why? If not, which ones are closest to being equally spaced?

Q35.4 In a two-slit interference pattern on a distant screen, are the bright fringes midway between the dark fringes? Is this ever a good approximation?

Q35.5 Would the headlights of a distant car form a two-source interference pattern? If so, how might it be observed? If not, why not?

Q35.6 The two sources S_1 and S_2 shown in Fig. 35.3 emit waves of the same wavelength λ and are in phase with each other. Suppose S_1 is a weaker source, so that the waves emitted by S_1 have half the amplitude of the waves emitted by S_2. How would this affect the positions of the antinodal lines and nodal lines? Would there be total reinforcement at points on the antinodal curves? Would there be total cancellation at points on the nodal curves? Explain your answers.

Q35.7 Could the Young two-slit interference experiment be performed with gamma rays? If not, why not? If so, discuss differences in the experimental design compared to the experiment with visible light.

Q35.8 Coherent red light illuminates two narrow slits that are 25 cm apart. Will a two-slit interference pattern be observed when the light from the slits falls on a screen? Explain.

Q35.9 Coherent light with wavelength λ falls on two narrow slits separated by a distance d. If d is less than some minimum value, no dark fringes are observed. Explain. In terms of λ, what is this minimum value of d?

Q35.10 A fellow student, who values memorizing equations above understanding them, combines Eqs. (35.4) and (35.13) to "prove" that ϕ can *only* equal $2\pi m$. How would you explain to this student that ϕ can have values other than $2\pi m$?

Q35.11 If the monochromatic light shown in Fig. 35.5a were replaced by white light, would a two-slit interference pattern be seen on the screen? Explain.

Q35.12 In using the superposition principle to calculate intensities in interference patterns, could you add the intensities of the waves instead of their amplitudes? Explain.

Q35.13 A glass windowpane with a thin film of water on it reflects less than when it is perfectly dry. Why?

Q35.14 A *very* thin soap film $(n = 1.33)$, whose thickness is much less than a wavelength of visible light, looks black; it appears to reflect no light at all. Why? By contrast, an equally thin layer of soapy water $(n = 1.33)$ on glass $(n = 1.50)$ appears quite shiny. Why is there a difference?

Q35.15 Interference can occur in thin films. Why is it important that the films be *thin*? Why don't you get these effects with a relatively *thick* film? Where should you put the dividing line between "thin" and "thick"? Explain your reasoning.

Q35.16 If we shine white light on an air wedge like that shown in Fig. 35.12, the colors that are weak in the light *reflected* from any point along the wedge are strong in the light *transmitted* through the wedge. Explain why this should be so.

Q35.17 Monochromatic light is directed at normal incidence on a thin film. There is destructive interference for the reflected light, so the intensity of the reflected light is very low. What happened to the energy of the incident light?

Q35.18 When a thin oil film spreads out on a puddle of water, the thinnest part of the film looks dark in the resulting interference pattern. What does this tell you about the relative magnitudes of the refractive indexes of oil and water?

EXERCISES

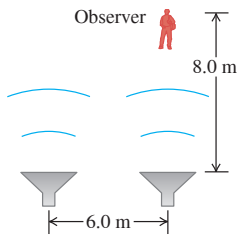
Section 35.1 Interference and Coherent Sources

35.1 •• A radio transmitting station operating at a frequency of 120 MHz has two identical antennas that radiate in phase. Antenna B is 9.00 m to the right of antenna A. Consider point P between the antennas and along the line connecting them, a horizontal distance x to the right of antenna A. For what values of x will constructive interference occur at point P?

35.2 • **Radio Interference.** Two radio antennas A and B radiate in phase. Antenna B is 110 m to the right of antenna A. Consider point Q along the extension of the line connecting the antennas, a horizontal distance of 30 m to the right of antenna B. The frequency, and hence the wavelength, of the emitted waves can be varied. (a) What is the longest wavelength for which there will be destructive interference at point Q? (b) What is the longest wavelength for which there will be constructive interference at point Q?

35.3 • Two speakers, emitting identical sound waves of wavelength 2.0 m in phase with each other, and an observer are located as shown in **Fig. E35.3**. (a) At the observer's location, what is the path difference for waves from the two speakers? (b) Will the sound waves interfere constructively or destructively at the observer's location—or something in between constructive and destructive? (c) Suppose the observer now increases her distance from the closest speaker to 17.0 m, staying directly in front of the same speaker as initially. Answer the questions of parts (a) and (b) for this new situation.

Figure **E35.3**



35.4 • Two light sources can be adjusted to emit monochromatic light of any visible wavelength. The two sources are coherent, 2.04 μm apart, and in line with an observer, so that one source is 2.04 μm farther from the observer than the other. (a) For what visible wavelengths (380 to 750 nm) will the observer see the brightest light, owing to constructive interference? (b) How would your answers to part (a) be affected if the two sources were not in line with the observer, but were still arranged so that one source is 2.04 μm farther away from the observer than the other? (c) For what visible wavelengths will there be *destructive* interference at the location of the observer?

35.5 •• Antenna B is 40.0 m to the right of antenna A. The two antennas emit electromagnetic waves that are in phase and have wavelength 7.00 m. (a) At how many points along the line connecting A and B is the interference constructive? (b) What is the smallest distance to the right of antenna A for which is there a point of constructive interference?

Section 35.2 Two-Source Interference of Light

35.6 •• Coherent light of wavelength 500 nm is incident on two very narrow and closely spaced slits. The interference pattern is observed on a very tall screen that is 2.00 m from the slits. Near the center of the screen the separation between two adjacent interference maxima is 3.53 cm. What is the distance on the screen between the $m = 49$ and $m = 50$ maxima?

35.7 • Young's experiment is performed with light from excited helium atoms ($\lambda = 502$ nm). Fringes are measured carefully on a screen 1.20 m away from the double slit, and the center of the 20th fringe (not counting the central bright fringe) is found to be 10.6 mm from the center of the central bright fringe. What is the separation of the two slits?

35.8 •• Coherent light with wavelength 450 nm falls on a pair of slits. On a screen 1.80 m away, the distance between dark fringes is 3.90 mm. What is the slit separation?

35.9 •• Two slits spaced 0.450 mm apart are placed 78.5 cm from a screen. What is the distance between the second and third dark lines of the interference pattern on the screen when the slits are illuminated with coherent light with a wavelength of 490 nm?

35.10 •• Coherent light of frequency f travels in air and is incident on two narrow slits. The interference pattern is observed on a distant screen that is directly opposite the slits. The frequency of light f can be varied. For $f = 5.60 \times 10^{12}$ Hz there is an interference maximum for $\theta = 60.0°$. The next higher frequency for which there is an interference maximum at this angle is 7.47×10^{12} Hz. What is the separation d between the two slits?

35.11 •• Two thin parallel slits that are 0.0116 mm apart are illuminated by a laser beam of wavelength 585 nm. (a) On a very large distant screen, what is the *total* number of bright fringes (those indicating complete constructive interference), including the central fringe and those on both sides of it? Solve this problem without calculating all the angles! (*Hint:* What is the largest that $\sin\theta$ can be? What does this tell you is the largest value of m?) (b) At what angle, relative to the original direction of the beam, will the fringe that is most distant from the central bright fringe occur?

35.12 • Coherent light with wavelength 400 nm passes through two very narrow slits that are separated by 0.200 mm, and the interference pattern is observed on a screen 4.00 m from the slits. (a) What is the width (in mm) of the central interference maximum? (b) What is the width of the first-order bright fringe?

35.13 •• Two very narrow slits are spaced 1.80 μm apart and are placed 35.0 cm from a screen. What is the distance between the first and second dark lines of the interference pattern when the slits are illuminated with coherent light with $\lambda = 550$ nm? (*Hint:* The angle θ in Eq. (35.5) is *not* small.)

35.14 •• Coherent light that contains two wavelengths, 660 nm (red) and 470 nm (blue), passes through two narrow slits that are separated by 0.280 mm. Their interference pattern is observed on a screen 4.20 m from the slits. What is the distance on the screen between the first-order bright fringes for the two wavelengths?

35.15 •• Coherent light with wavelength 600 nm passes through two very narrow slits and the interference pattern is observed on a screen 3.00 m from the slits. The first-order bright fringe is at 4.84 mm from the center of the central bright fringe. For what wavelength of light will the first-order dark fringe be observed at this same point on the screen?

35.16 •• Coherent light of frequency 6.32×10^{14} Hz passes through two thin slits and falls on a screen 85.0 cm away. You observe that the third bright fringe occurs at ± 3.11 cm on either side of the central bright fringe. (a) How far apart are the two slits? (b) At what distance from the central bright fringe will the third dark fringe occur?

Section 35.3 Intensity in Interference Patterns

35.17 •• In a two-slit interference pattern, the intensity at the peak of the central maximum is I_0. (a) At a point in the pattern where the phase difference between the waves from the two slits is 65.0°, what is the intensity? (b) What is the path difference for 478 nm light from the two slits at a point where the phase difference is 65.0°?

35.18 • Two slits spaced 0.260 mm apart are 0.650 m from a screen and illuminated by coherent light of wavelength 630 nm. The intensity at the center of the central maximum ($\theta = 0°$) is I_0. What is the distance on the screen from the center of the central maximum (a) to the first minimum; (b) to the point where the intensity has fallen to $I_0/2$?

35.19 • Coherent light with wavelength 500 nm passes through narrow slits separated by 0.340 mm. At a distance from the slits large compared to their separation, what is the phase difference (in radians) in the light from the two slits at an angle of 23.0° from the centerline?

35.20 •• Two slits spaced 0.0720 mm apart are 0.800 m from a screen. Coherent light of wavelength λ passes through the two slits. In their interference pattern on the screen, the distance from the center of the central maximum to the first minimum is 3.00 mm. If the intensity at the peak of the central maximum is 0.0600 W/m^2, what is the intensity at points on the screen that are (a) 2.00 mm and (b) 1.50 mm from the center of the central maximum?

35.21 • Consider two antennas separated by 9.00 m that radiate in phase at 120 MHz, as described in Exercise 35.1. A receiver placed 150 m from both antennas measures an intensity I_0. The receiver is moved so that it is 1.8 m closer to one antenna than to the other. (a) What is the phase difference ϕ between the two radio waves produced by this path difference? (b) In terms of I_0, what is the intensity measured by the receiver at its new position?

Section 35.4 Interference in Thin Films

35.22 •• Two identical horizontal sheets of glass have a thin film of air of thickness t between them. The glass has refractive index 1.40. The thickness t of the air layer can be varied. Light with wavelength λ in air is at normal incidence onto the top of the air film. There is constructive interference between the light reflected at the top and bottom surfaces of the air film when its thickness is 650 nm. For the same wavelength of light the next larger thickness for which there is constructive interference is 910 nm. (a) What is the wavelength λ of the light when it is traveling in air? (b) What is the smallest thickness t of the air film for which there is constructive interference for this wavelength of light?

35.23 • What is the thinnest film of a coating with $n = 1.42$ on glass ($n = 1.52$) for which destructive interference of the red component (650 nm) of an incident white light beam in air can take place by reflection?

35.24 •• **Nonglare Glass.** When viewing a piece of art that is behind glass, one often is affected by the light that is reflected off the front of the glass (called *glare*), which can make it difficult to see the art clearly. One solution is to coat the outer surface of the glass with a film to cancel part of the glare. (a) If the glass has a refractive index of 1.62 and you use TiO_2, which has an index of refraction of 2.62, as the coating, what is the minimum film thickness that will cancel light of wavelength 505 nm? (b) If this coating is too thin to stand up to wear, what other thickness would also work? Find only the three thinnest ones.

35.25 •• A uniform film of TiO_2, 1036 nm thick and having index of refraction 2.62, is spread uniformly over the surface of crown glass of refractive index 1.52. Light of wavelength 520.0 nm falls at normal incidence onto the film from air. You want to increase the thickness of this film so that the reflected light cancels. (a) What is the *minimum* thickness of TiO_2 that you must *add* so the reflected light cancels as desired? (b) After you make the adjustment in part (a), what is the path difference between the light reflected off the top of the film and the light that cancels it after traveling through the film? Express your answer in (i) nanometers and (ii) wavelengths of the light in the TiO_2 film.

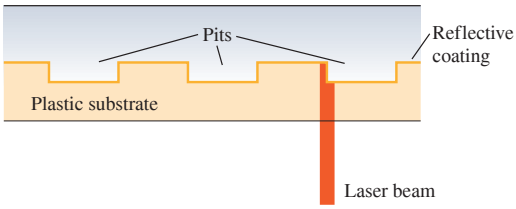
35.26 • A plastic film with index of refraction 1.70 is applied to the surface of a car window to increase the reflectivity and thus to keep the car's interior cooler. The window glass has index of refraction 1.52. (a) What minimum thickness is required if light of wavelength 550 nm in air reflected from the two sides of the film is to interfere constructively? (b) Coatings as thin as that calculated in part (a) are difficult to manufacture and install. What is the next greater thickness for which constructive interference will also occur?

35.27 • The walls of a soap bubble have about the same index of refraction as that of plain water, $n = 1.33$. There is air both inside and outside the bubble. (a) What wavelength (in air) of visible light is most strongly reflected from a point on a soap bubble where its wall is 290 nm thick? To what color does this correspond (see Fig. 32.4 and Table 32.1)? (b) Repeat part (a) for a wall thickness of 340 nm.

35.28 •• A researcher measures the thickness of a layer of benzene ($n = 1.50$) floating on water by shining monochromatic light onto the film and varying the wavelength of the light. She finds that light of wavelength 575 nm is reflected most strongly from the film. What does she calculate for the minimum thickness of the film?

35.29 •• **Compact Disc Player.** A compact disc (CD) is read from the bottom by a semiconductor laser with wavelength 790 nm passing through a plastic substrate of refractive index 1.8. When the beam encounters a pit, part of the beam is reflected from the pit and part from the flat region between the pits, so these two beams interfere with each other (**Fig. E35.29**). What must the minimum pit depth be so that the part of the beam reflected from a pit cancels the part of the beam reflected from the flat region? (It is this cancellation that allows the player to recognize the beginning and end of a pit.)

Figure **E35.29**



Section 35.5 The Michelson Interferometer

35.30 • Jan first uses a Michelson interferometer with the 606 nm light from a krypton-86 lamp. He displaces the movable mirror away from him, counting 818 fringes moving across a line in his field of view. Then Linda replaces the krypton lamp with filtered 502 nm light from a helium lamp and displaces the movable mirror toward her. She also counts 818 fringes, but they move across the line in her field of view opposite to the direction they moved for Jan. Assume that both Jan and Linda counted to 818 correctly. (a) What distance did each person move the mirror? (b) What is the resultant displacement of the mirror?

35.31 • How far must the mirror M_2 (see Fig. 35.19) of the Michelson interferometer be moved so that 1800 fringes of He-Ne laser light ($\lambda = 633$ nm) move across a line in the field of view?

PROBLEMS

35.32 •• The LIGO experiment, which historically detected gravitational waves for the first time in September 2015, uses a pair of highly sensitive Michelson interferometers. These have arms that are 4.00 km long and use powerful Nd:Yag lasers with 1064 nm wavelength. The beams traverse the arms both ways 280 times before recombining, which effectively lengthens the arm length to 1120 km. The devices are tuned so that the beams destructively interfere when they recombine if no gravitational wave is present. (a) The beam has a power of 100 kW, concentrated into an area of a square centimeter. Calculate the amplitude of the electric field in the beam. (b) LIGO can detect a gravitational wave that temporarily lengthens one arm by the minuscule amount of 10^{-18} m! When this happens, the beams combine with a phase difference of $\pi + \delta$. Estimate the shift δ in radians. Note that the phase difference accumulates during both traversals of each round trip. (c) Use Eq. (35.7) to estimate the sensitivity of the photodetector in terms of the minimal electric field strength needed to detect a gravitational wave.

35.33 •• One round face of a 3.50 m, solid, cylindrical plastic pipe is covered with a thin black coating that completely blocks light. The opposite face is covered with a fluorescent coating that glows when it is struck by light. Two straight, thin, parallel scratches, 0.215 mm apart, are made in the center of the black face. When laser light of wavelength 632.8 nm shines through the slits perpendicular to the black face, you find that the central bright fringe on the opposite face is 5.62 mm wide, measured between the dark fringes that border it on either side. What is the index of refraction of the plastic?

35.34 ••• Newton's rings are visible when a planoconvex lens is placed on a flat glass surface. For a particular lens with an index of refraction of $n = 1.46$ and a glass plate with an index of $n = 1.85$, the diameter of the third bright ring is 0.890 mm. If water ($n = 1.33$) now fills the space between the lens and the glass plate, what is the new diameter of this ring? Assume the radius of curvature of the lens is much greater than the wavelength of the light.

35.35 • BIO **Coating Eyeglass Lenses.** Eyeglass lenses can be coated on the *inner* surfaces to reduce the reflection of stray light to the eye. If the lenses are medium flint glass of refractive index 1.62 and the coating is fluorite of refractive index 1.432, (a) what minimum thickness of film is needed on the lenses to cancel light of wavelength 550 nm reflected toward the eye at normal incidence? (b) Will any other wavelengths of visible light be cancelled or enhanced in the reflected light?

35.36 •• BIO **Sensitive Eyes.** After an eye examination, you put some eyedrops on your sensitive eyes. The cornea (the front part of the eye) has an index of refraction of 1.38, while the eyedrops have a refractive index of 1.45. After you put in the drops, your friends notice that your eyes look red, because red light of wavelength 600 nm has been reinforced in the reflected light. (a) What is the minimum thickness of the film of eyedrops on your cornea? (b) Will any other wavelengths of visible light be reinforced in the reflected light? Will any be cancelled? (c) Suppose you had contact lenses, so that the eyedrops went on them instead of on your corneas. If the refractive index of the lens material is 1.50 and the layer of eyedrops has the same thickness as in part (a), what wavelengths of visible light will be reinforced? What wavelengths will be cancelled?

35.37 •• Two flat plates of glass with parallel faces are on a table, one plate on the other. Each plate is 11.0 cm long and has a refractive index of 1.55. A very thin sheet of metal foil is inserted under the end of the upper plate to raise it slightly at that end, in a manner similar to that discussed in Example 35.4. When you view the glass plates from above with reflected white light, you observe that, at 1.15 mm from the line where the sheets are in contact, the violet light of wavelength 400.0 nm is enhanced in this reflected light, but no visible light is enhanced closer to the line of contact. (a) How far from the line of contact will green light (of wavelength 550.0 nm) and orange light (of wavelength 600.0 nm) first be enhanced? (b) How far from the line of contact will the violet, green, and orange light again be enhanced in the reflected light? (c) How thick is the metal foil holding the ends of the plates apart?

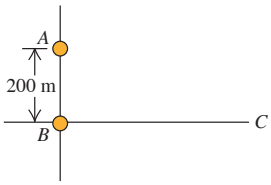
35.38 •• A typical red laser pointer has a wavelength of 650 nm. Suppose we wanted to test the wave nature of light by carefully cutting two parallel slits in a dark plastic sheet. We would then shine the laser through the slits onto a wall located 1 m beyond the sheet. (a) Determine the slit spacing needed so that the bright spots on the wall would be discernible with 1 cm spacing. (b) Is this a feasible "home experiment"? Is it possible to cut slits with that separation using typical household tools? (c) Suppose we wanted to test the wave nature of sound in a similar manner, by placing two small speakers driven in-phase near each other, separated by 40 cm, both facing a wall 2 m distant. If we used a 1.0 kHz tone, determine the distance between points along the wall that would exhibit enhanced sound. (d) Suppose we wanted the sound-enhanced points to be separated by only 1.75 m to render this experiment feasible. Estimate an audible frequency f and use it to determine a speaker separation distance d that would accomplish this. (e) Is this a feasible "home experiment"?

35.39 ••• Suppose you illuminate two thin slits by monochromatic coherent light in air and find that they produce their first interference *minima* at $\pm 35.09°$ on either side of the central bright spot. You then immerse these slits in a transparent liquid and illuminate them with the same light. Now you find that the first minima occur at $\pm 19.36°$ instead. What is the index of refraction of this liquid?

35.40 •• **CP CALC** A very thin sheet of brass contains two thin parallel slits. When a laser beam shines on these slits at normal incidence and room temperature (20.0°C), the first interference dark fringes occur at $\pm 26.6°$ from the original direction of the laser beam when viewed from some distance. If this sheet is now slowly heated to 135°C, by how many degrees do these dark fringes change position? Do they move closer together or farther apart? See Table 17.1 for pertinent information, and ignore any effects that might occur due to a change in the thickness of the slits. (*Hint:* Thermal expansion normally produces very small changes in length, so you can use differentials to find the change in the angle.)

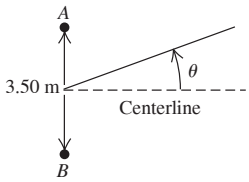
35.41 •• Two radio antennas radiating in phase are located at points A and B, 200 m apart (**Fig. P35.41**). The radio waves have a frequency of 5.80 MHz. A radio receiver is moved out from point B along a line perpendicular to the line connecting A and B (line BC shown in Fig. P35.41). At what distances from B will there be *destructive* interference? (*Note:* The distance of the receiver from the sources is not large in comparison to the separation of the sources, so Eq. (35.5) does not apply.)

Figure **P35.41**



35.42 •• Two speakers A and B are 3.50 m apart, and each one is emitting a frequency of 444 Hz. However, because of signal delays in the cables, speaker A is one-fourth of a period *ahead* of speaker B. For points far from the speakers, find all the angles relative to the centerline (**Fig. P35.42**) at which the sound from these speakers cancels. Include angles on *both* sides of the centerline. The speed of sound is 340 m/s.

Figure **P35.42**



35.43 •• **CP** A thin uniform film of refractive index 1.750 is placed on a sheet of glass of refractive index 1.50. At room temperature (20.0°C), this film is just thick enough for light with wavelength 582.4 nm reflected off the top of the film to be cancelled by light reflected from the top of the glass. After the glass is placed in an oven and slowly heated to 170°C, you find that the film cancels reflected light with wavelength 588.5 nm. What is the coefficient of linear expansion of the film? (Ignore any changes in the refractive index of the film due to the temperature change.)

35.44 ••• **GPS Transmission.** The GPS (Global Positioning System) satellites are approximately 5.18 m across and transmit two low-power signals, one of which is at 1575.42 MHz (in the UHF band). In a series of laboratory tests on the satellite, you put two 1575.42 MHz UHF transmitters at opposite ends of the satellite. These broadcast in phase uniformly in all directions. You measure the intensity at points on a circle that is several hundred meters in radius and centered on the satellite. You measure angles on this circle relative to a point that lies along the centerline of the satellite (that is, the perpendicular bisector of a line that extends from one transmitter to the other). At this point on the circle, the measured intensity is 2.00 W/m^2. (a) At how many other angles in the range $0° < \theta < 90°$ is the intensity also 2.00 W/m^2? (b) Find the four smallest angles in the range $0° < \theta < 90°$ for which the intensity is 2.00 W/m^2. (c) What is the intensity at a point on the circle at an angle of 4.65° from the centerline?

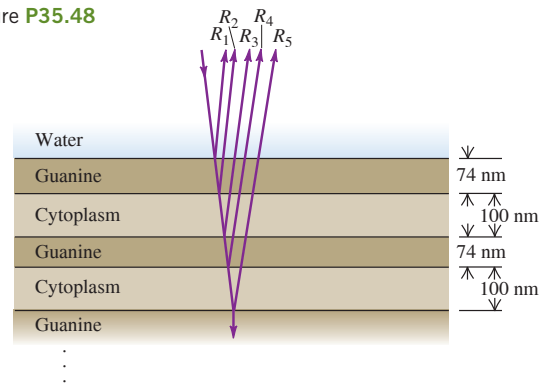
35.45 •• White light reflects at normal incidence from the top and bottom surfaces of a glass plate ($n = 1.52$). There is air above and below the plate. Constructive interference is observed for light whose wavelength in air is 477.0 nm. What is the thickness of the plate if the next longer wavelength for which there is constructive interference is 540.6 nm?

35.46 •• Laser light of wavelength 510 nm is traveling in air and shines at normal incidence onto the flat end of a transparent plastic rod that has $n = 1.30$. The end of the rod has a thin coating of a transparent material that has refractive index 1.65. What is the minimum (nonzero) thickness of the coating (a) for which there is maximum transmission of the light into the rod; (b) for which transmission into the rod is minimized?

35.47 •• Red light with wavelength 700 nm is passed through a two-slit apparatus. At the same time, monochromatic visible light with another wavelength passes through the same apparatus. As a result, most of the pattern that appears on the screen is a mixture of two colors; however, the center of the third bright fringe ($m = 3$) of the red light appears pure red, with none of the other color. What are the possible wavelengths of the second type of visible light? Do you need to know the slit spacing to answer this question? Why or why not?

35.48 •• **BIO Reflective Coatings and Herring.** Herring and related fish have a brilliant silvery appearance that camouflages them while they are swimming in a sunlit ocean. The silveriness is due to *platelets* attached to the surfaces of these fish. Each platelet is made up of several alternating layers of crystalline guanine ($n = 1.80$) and of cytoplasm ($n = 1.333$, the same as water), with a guanine layer on the outside in contact with the surrounding water (**Fig. P35.48**). In one typical platelet, the guanine layers are 74 nm thick and the cytoplasm layers are 100 nm thick. (a) For light striking the platelet surface at normal incidence, for which vacuum wavelengths of visible light will all of the reflections R_1, R_2, R_3, R_4, and R_5, shown in Fig. P35.48, be approximately in phase? If white light is shone on this platelet, what color will be most strongly reflected (see Fig. 32.4)? The surface of a herring has very many platelets side by side with layers of different thickness, so that *all* visible wavelengths are reflected. (b) Explain why such a "stack" of layers is more reflective than a single layer of guanine with cytoplasm underneath. (A stack of five guanine layers separated by cytoplasm layers reflects more than 80% of incident light at the wavelength for which it is "tuned.") (c) The color that is most strongly reflected from a platelet depends on the angle at which it is viewed. Explain why this should be so. (You can see these changes in color by examining a herring from different angles. Most of the platelets on these fish are oriented in the same way, so that they are vertical when the fish is swimming.)

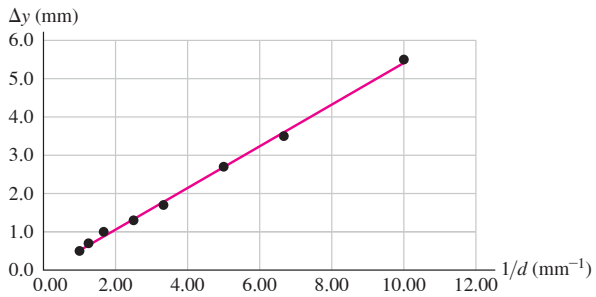
Figure **P35.48**



35.49 •• After a laser beam passes through two thin parallel slits, the first completely dark fringes occur at $\pm 17.0°$ with the original direction of the beam, as viewed on a screen far from the slits. (a) What is the ratio of the distance between the slits to the wavelength of the light illuminating the slits? (b) What is the smallest angle, relative to the original direction of the laser beam, at which the intensity of the light is $\frac{1}{10}$ the maximum intensity on the screen?

35.50 •• **DATA** In your summer job at an optics company, you are asked to measure the wavelength λ of the light that is produced by a laser. To do so, you pass the laser light through two narrow slits that are separated by a distance d. You observe the interference pattern on a screen that is 0.900 m from the slits and measure the separation Δy between adjacent bright fringes in the portion of the pattern that is near the center of the screen. Using a microscope, you measure d. But both Δy and d are small and difficult to measure accurately, so you repeat the measurements for several pairs of slits, each with a different value of d. Your results are shown in **Fig. P35.50**, where you have plotted Δy versus $1/d$. The line in the graph is the best-fit straight line for the data. (a) Explain why the data points plotted this way fall close to a straight line. (b) Use Fig. P35.50 to calculate λ.

Figure **P35.50**

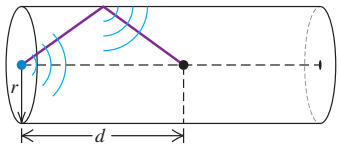


35.51 •• **DATA** Short-wave radio antennas A and B are connected to the same transmitter and emit coherent waves in phase and with the same frequency f. You must determine the value of f and the placement of the antennas that produce a maximum intensity through constructive interference at a receiving antenna that is located at point P, which is at the corner of your garage. First you place antenna A at a point 240.0 m due east of P. Next you place antenna B on the line that connects A and P, a distance x due east of P, where $x < 240.0$ m. Then you measure that a maximum in the total intensity from the two antennas occurs when $x = 210.0$ m, 216.0 m, and 222.0 m. You don't investigate smaller or larger values of x. (Treat the antennas as point sources.) (a) What is the frequency f of the waves that are emitted by the antennas? (b) What is the greatest value of x, with $x < 240.0$ m, for which the interference at P is destructive?

35.52 •• **DATA** In your research lab, a very thin, flat piece of glass with refractive index 1.40 and uniform thickness covers the opening of a chamber that holds a gas sample. The refractive indexes of the gases on either side of the glass are very close to unity. To determine the thickness of the glass, you shine coherent light of wavelength λ_0 in vacuum at normal incidence onto the surface of the glass. When $\lambda_0 = 496$ nm, constructive interference occurs for light that is reflected at the two surfaces of the glass. You find that the next shorter wavelength in vacuum for which there is constructive interference is 386 nm. (a) Use these measurements to calculate the thickness of the glass. (b) What is the longest wavelength in vacuum for which there is constructive interference for the reflected light?

35.53 •• **CP** An acoustic waveguide consists of a long cylindrical tube with radius r designed to channel sound waves, as shown in **Fig. P35.53**. A tone with frequency f is emitted from a small source at the center of one end of this tube. Depending on the radius of the tube and the frequency of the tone, pressure nodes can develop along the tube axis where rays reflected from the periphery constructively interfere with direct rays. (a) For sound waves with wavelength λ, what is the minimum tube radius for which at least one such node exists? (b) The tube has radius 25.0 cm and the temperature is 20°C. If the tone has frequency 2.50 kHz, how many nodes exist? (c) At what distance d are these nodes located? (d) If the tube were filled with helium rather than air, how many nodes would exist, and at what value of d?

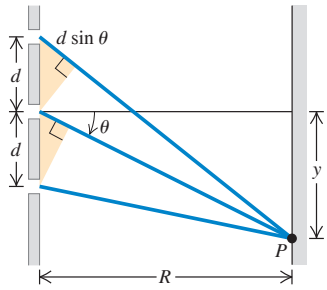
Figure **P35.53**



CHALLENGE PROBLEMS

35.54 ••• **Three-Slit Interference.** Monochromatic light with wavelength λ is incident on a screen with three narrow slits with separation d, as shown in **Fig. P35.54**. Light from the middle slit reaches point P with electric field $E\cos(\omega t)$. From the small-angle approximation, light from the upper and lower slits reaches point P with electric fields $E\cos(\omega t + \phi)$ and $E\cos(\omega t - \phi)$, respectively, where $\phi = (2\pi d \sin\theta)/\lambda$ is the phase lag and phase lead associated with the different path lengths. (a) Using either a phasor analysis similar to Fig. 35.9 or trigonometric identities, determine the electric-field amplitude E_P associated with the net field at point P, in terms of ϕ. (b) Determine the intensity I at point P in terms of the maximum net intensity I_0 and the phase angle ϕ. (c) There are two sets of relative maxima: one with intensity I_0 when $\phi = 2\pi m$, where m is an integer, and another with a smaller intensity at other values of ϕ. What values of ϕ exhibit the "lesser" maxima? (d) What is the intensity at the lesser maxima, in terms of I_0? (e) What values of ϕ correspond to the dark fringes closest to the center? (f) If the incident light has wavelength 650 nm, the slits are separated by $d = 0.200$ mm, and the distance to the far screen is $R = 1.00$ m, what is the distance from the central maximum to the first lesser maximum? (g) What is the distance from the central maximum to the closest absolute maximum?

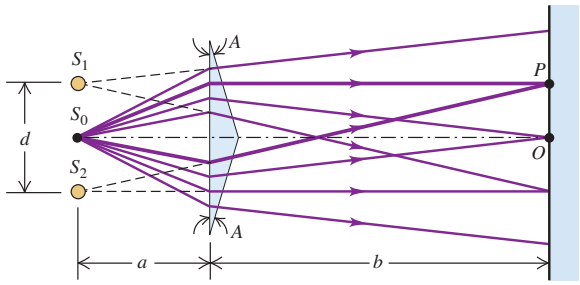
Figure **P35.54**



35.55 ••• **CP** The index of refraction of a glass rod is 1.48 at $T = 20.0$°C and varies linearly with temperature, with a coefficient of 2.50×10^{-5}/°C. The coefficient of linear expansion of the glass is 5.00×10^{-6}/°C. At 20.0°C the length of the rod is 3.00 cm. A Michelson interferometer has this glass rod in one arm, and the rod is being heated so that its temperature increases at a rate of 5.00 °C/min. The light source has wavelength $\lambda = 589$ nm, and the rod initially is at $T = 20.0$°C. How many fringes cross the field of view each minute?

35.56 ••• CP **Figure P35.56** shows an interferometer known as *Fresnel's biprism*. The magnitude of the prism angle A is extremely small. (a) If S_0 is a very narrow source slit, show that the separation of the two virtual coherent sources S_1 and S_2 is given by $d = 2aA(n - 1)$, where n is the index of refraction of the material of the prism. (b) Calculate the spacing of the fringes of green light with wavelength 500 nm on a screen 2.00 m from the biprism. Take $a = 0.200$ m, $A = 3.50$ mrad, and $n = 1.50$.

Figure P35.56



MCAT-STYLE PASSAGE PROBLEMS

Interference and Sound Waves. Interference occurs with not only light waves but also all frequencies of electromagnetic waves and all other types of waves, such as sound and water waves. Suppose that your physics professor sets up two sound speakers in the front of your classroom and uses an electronic oscillator to produce sound waves of a single frequency. When she turns the oscillator on (take this to be its original setting), you and many students hear a loud tone while other students hear nothing. (The speed of sound in air is 340 m/s.)

35.57 The professor then adjusts the apparatus. The frequency that you hear does not change, but the loudness decreases. Now all of your fellow students can hear the tone. What did the professor do? (a) She turned off the oscillator. (b) She turned down the volume of the speakers. (c) She changed the phase relationship of the speakers. (d) She disconnected one speaker.

35.58 The professor returns the apparatus to the original setting. She then adjusts the speakers again. All of the students who had heard nothing originally now hear a loud tone, while you and the others who had originally heard the loud tone hear nothing. What did the professor do? (a) She turned off the oscillator. (b) She turned down the volume of the speakers. (c) She changed the phase relationship of the speakers. (d) She disconnected one speaker.

35.59 The professor again returns the apparatus to its original setting, so you again hear the original loud tone. She then slowly moves one speaker away from you until it reaches a point at which you can no longer hear the tone. If she has moved the speaker by 0.34 m (farther from you), what is the frequency of the tone? (a) 1000 Hz; (b) 2000 Hz; (c) 500 Hz; (d) 250 Hz.

35.60 The professor once again returns the apparatus to its original setting, but now she adjusts the oscillator to produce sound waves of half the original frequency. What happens? (a) The students who originally heard a loud tone again hear a loud tone, and the students who originally heard nothing still hear nothing. (b) The students who originally heard a loud tone now hear nothing, and the students who originally heard nothing now hear a loud tone. (c) Some of the students who originally heard a loud tone again hear a loud tone, but others in that group now hear nothing. (d) Among the students who originally heard nothing, some still hear nothing but others now hear a loud tone.

ANSWERS

Chapter Opening Question ?

(v) The colors appear as a result of constructive interference between light waves reflected from the upper and lower surfaces of the oil film. The wavelength of light for which the most constructive interference occurs at a point, and hence the color that appears the brightest at that point, depends on (1) the thickness of the film (which determines the path difference between light waves that reflect off the two surfaces), (2) the oil's index of refraction (which gives the wavelength of light in the oil a different value than in air), and (3) the index of refraction of the material below the oil (which determines whether the wave that reflects from the lower surface undergoes a half-cycle phase shift). (See Examples 35.4, 35.5, and 35.6 in Section 35.4.)

Key Example VARIATION Problems

VP35.2.1 (a) 684 nm (b) 17.1 mm
VP35.2.2 (a) 0°, 20.7°, 44.9° (b) 10.2°, 31.9°, 61.9°
VP35.2.3 (a) 0.472 mm (b) 6.88 mm
VP35.2.4 (a) 657 nm (b) 0.254°
VP35.3.1 (a) 0.0192 W/m^2 (b) 5.58°
VP35.3.2 (a) 0.0173 W/m^2 (b) 1.66 mm
VP35.3.3 (a) 3.63 m (b) 9.14×10^{-4} W/m^2
VP35.3.4 (a) 573 nm (b) 8.22 mm
VP35.6.1 (a) 0.0174 mm (b) 0.956 mm
VP35.6.2 (a) constructive: 584 nm; destructive: none (b) constructive: none; destructive: 426 nm
VP35.6.3 (a) constructive: none; destructive: 688 nm (b) constructive: 468 nm; destructive: none
VP35.6.4 (a) 88.3 nm (b) 265 nm

Bridging Problem

(a) 441 nm; violet
(b) 551 nm